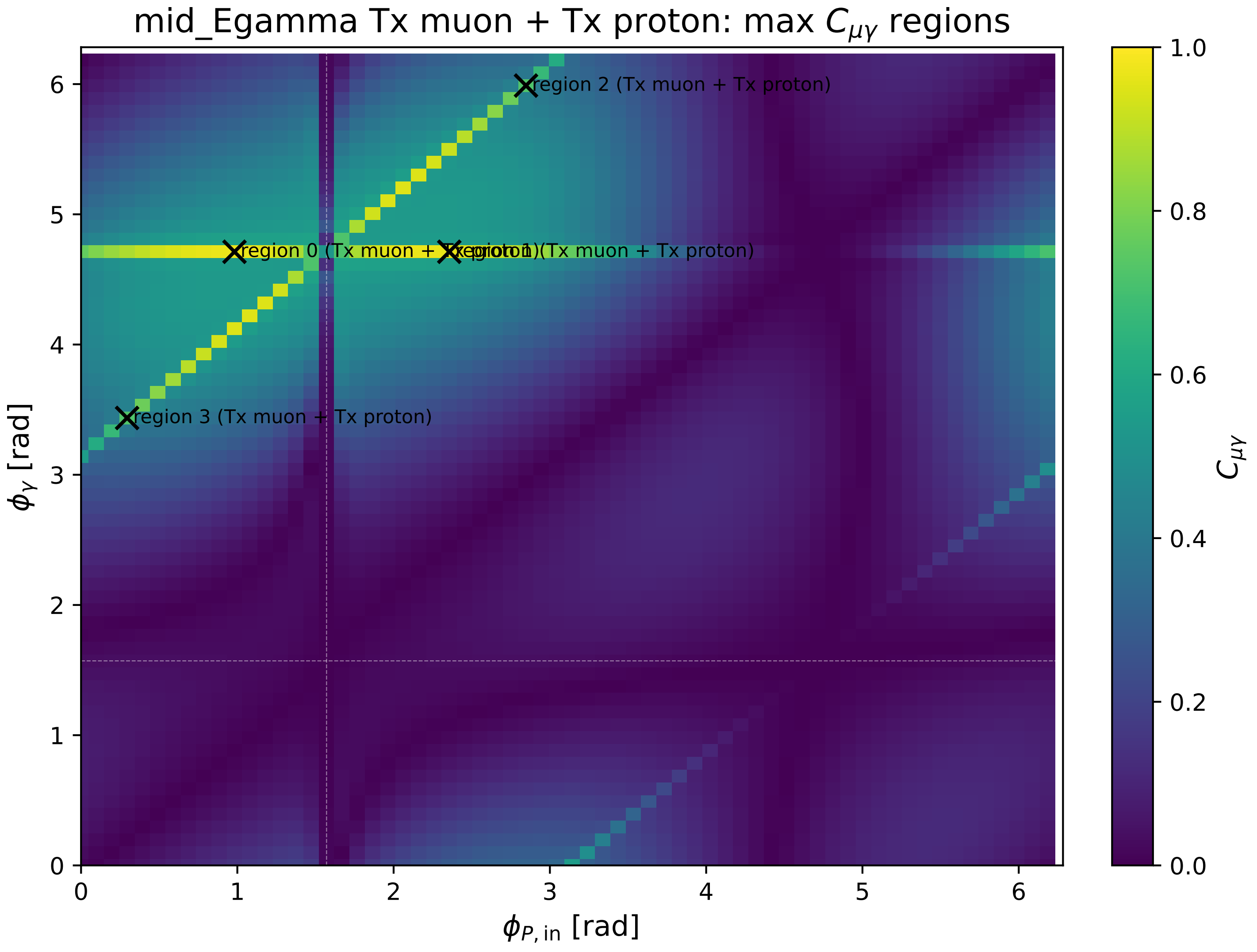
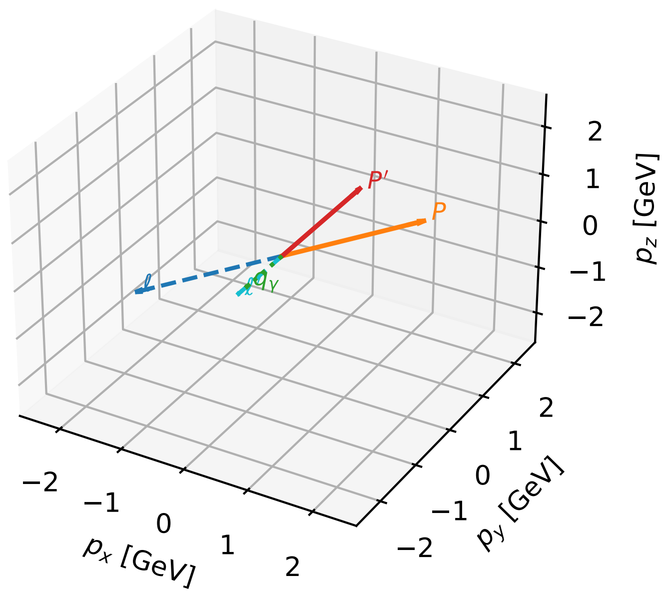

mid_Egamma Tx muon + Tx proton: max $C_{\mu\gamma}$ regions
region 0 (Tx muon + Tx proton)
region 1 (Tx muon + Tx proton)
region 2 (Tx muon + Tx proton)
region 3 (Tx muon + Tx proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Tx muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration




```
c_mu_gamma_mid_egamma_txtx_region_0  C_{mu\gamma}=0.968874
region: mid_Egamma_TxTx_region_0  final-pair delta_xy=0 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
```
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.22204

θ_{in}=1.57079, ϕ_ℓ=4.12334, ϕ_P=0.981748

E_γ=1, ϕ_γ=4.71239

Q^2=0.919807, x_B=0.0902858, t=-1.66503, W^2=10.1478, y=0.493954

$\theta(\ell', \gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:

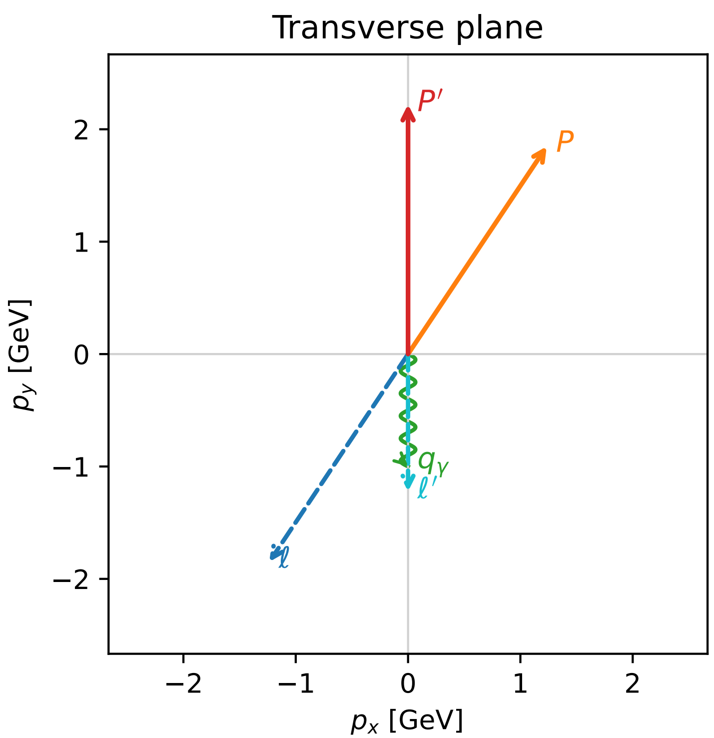
ℓ E= 2.2256 p_x= -1.2351 p_y= -1.8484 p_z= -0.0000

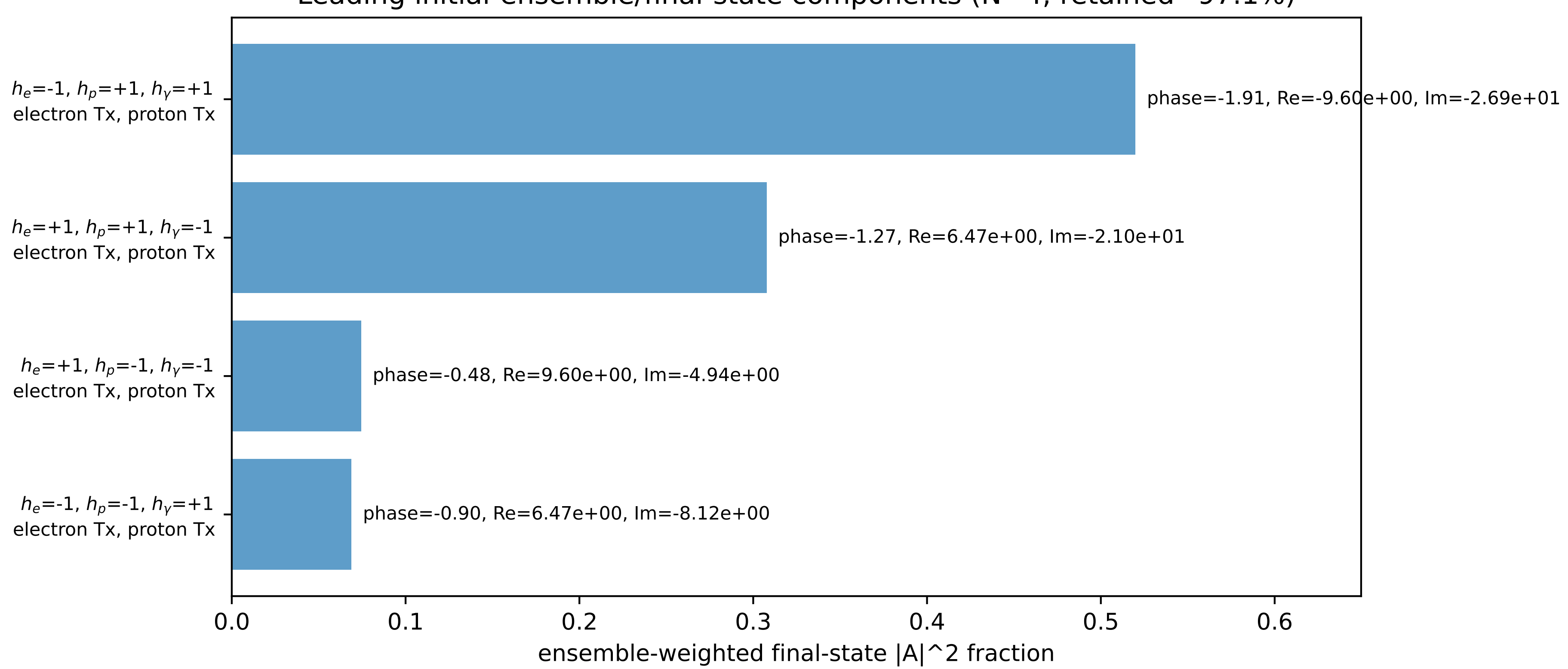
P E= 2.4129 p_x= 1.2351 p_y= 1.8484 p_z= 0.0000

ℓ' E= 1.2266 p_x= 0.0000 p_y= -1.2220 p_z= 0.0000

P' E= 2.4119 p_x= 0.0000 p_y= 2.2220 p_z= 0.0000

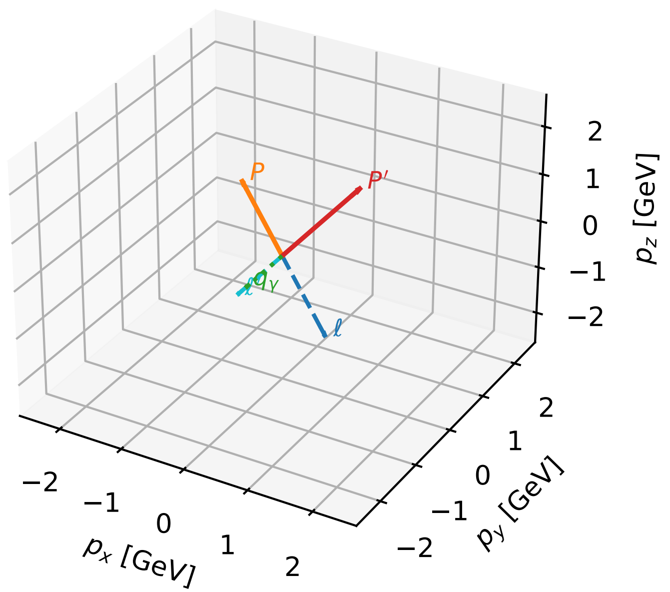
q_γ E= 1.0000 p_x= -0.0000 p_y= -1.0000 p_z= 0.0000





Tx muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration

3D momenta

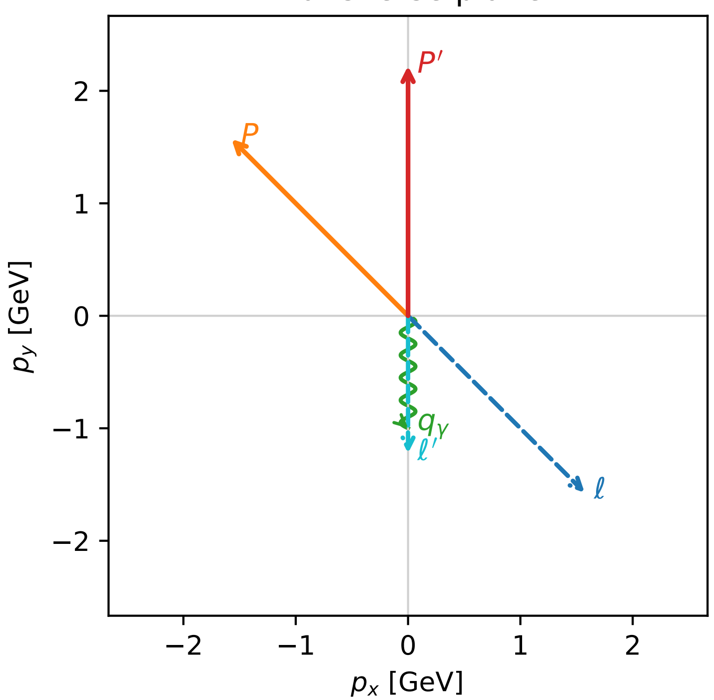



```
c_mu_gamma_mid_egamma_txtx_region_1  C_{mu\gamma}=0.963566
region: mid_Egamma_TxTx_region_1  final-pair delta_xy=0 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
```
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

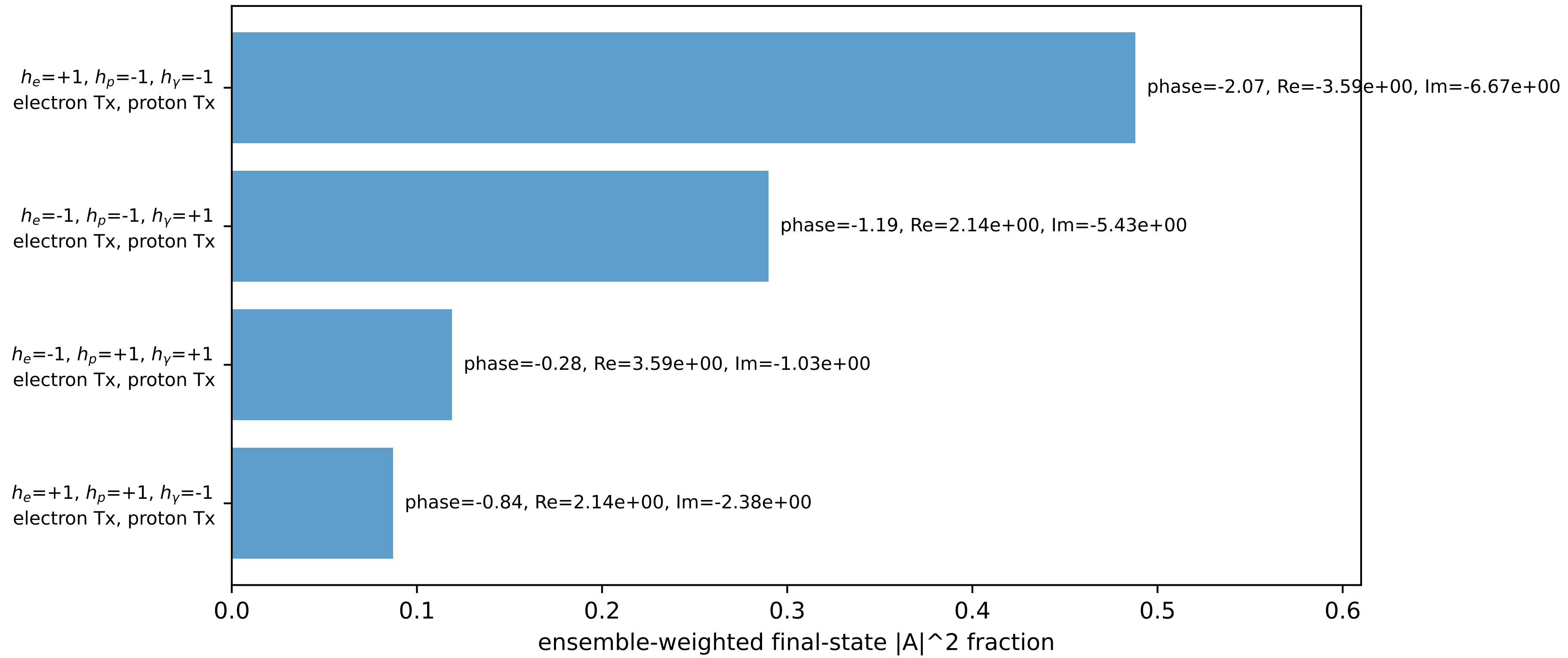
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.22204
θ_{in}=1.57079, ϕ_ℓ=5.49779, ϕ_P=2.35619
E_γ=1, ϕ_γ=4.71239
Q^2=1.59553, x_B=0.146871, t=-2.8937, W^2=10.1478, y=0.526717
$\theta(\ell', \gamma)$=0 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 1.5720 p_y= -1.5720 p_z= -0.0000
P E= 2.4129 p_x= -1.5720 p_y= 1.5720 p_z= 0.0000
ℓ' E= 1.2266 p_x= 0.0000 p_y= -1.2220 p_z= 0.0000
P' E= 2.4119 p_x= 0.0000 p_y= 2.2220 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.0000 p_y= -1.0000 p_z= 0.0000

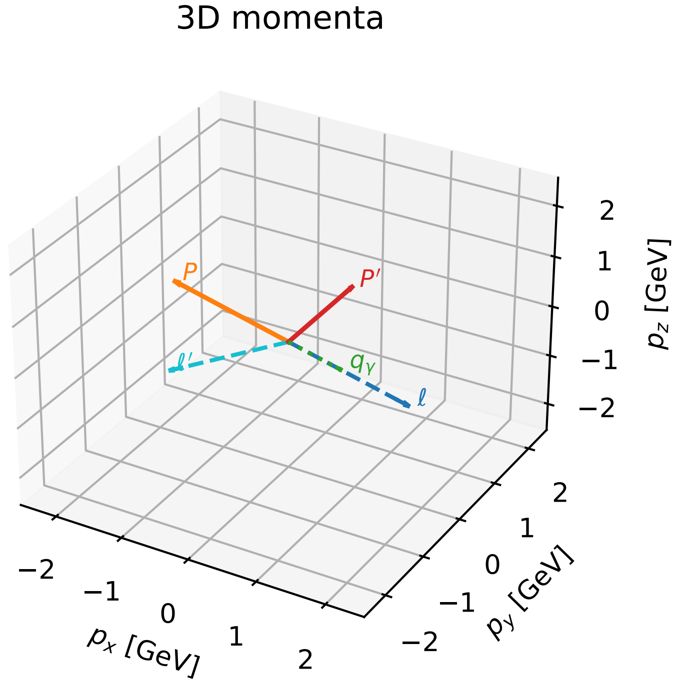
Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=98.4%)



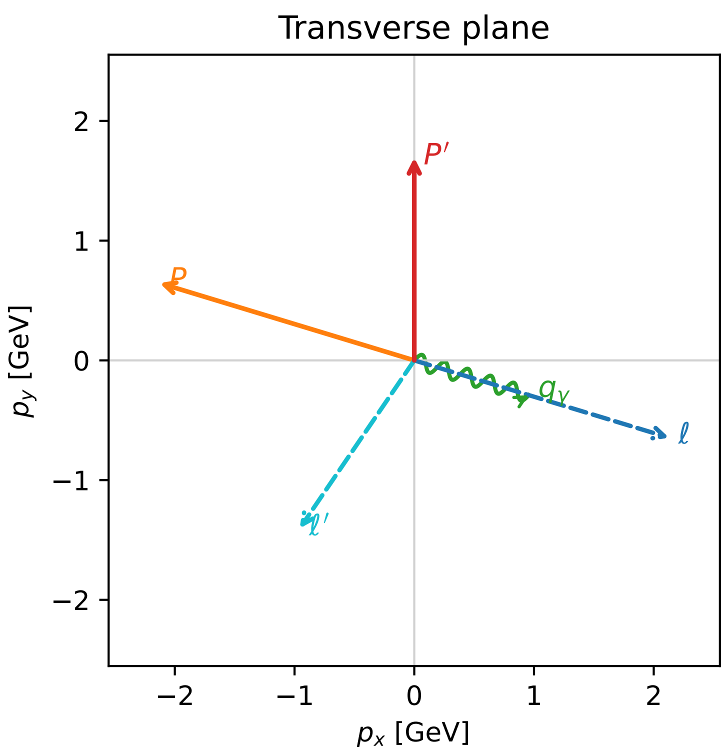
Tx muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration

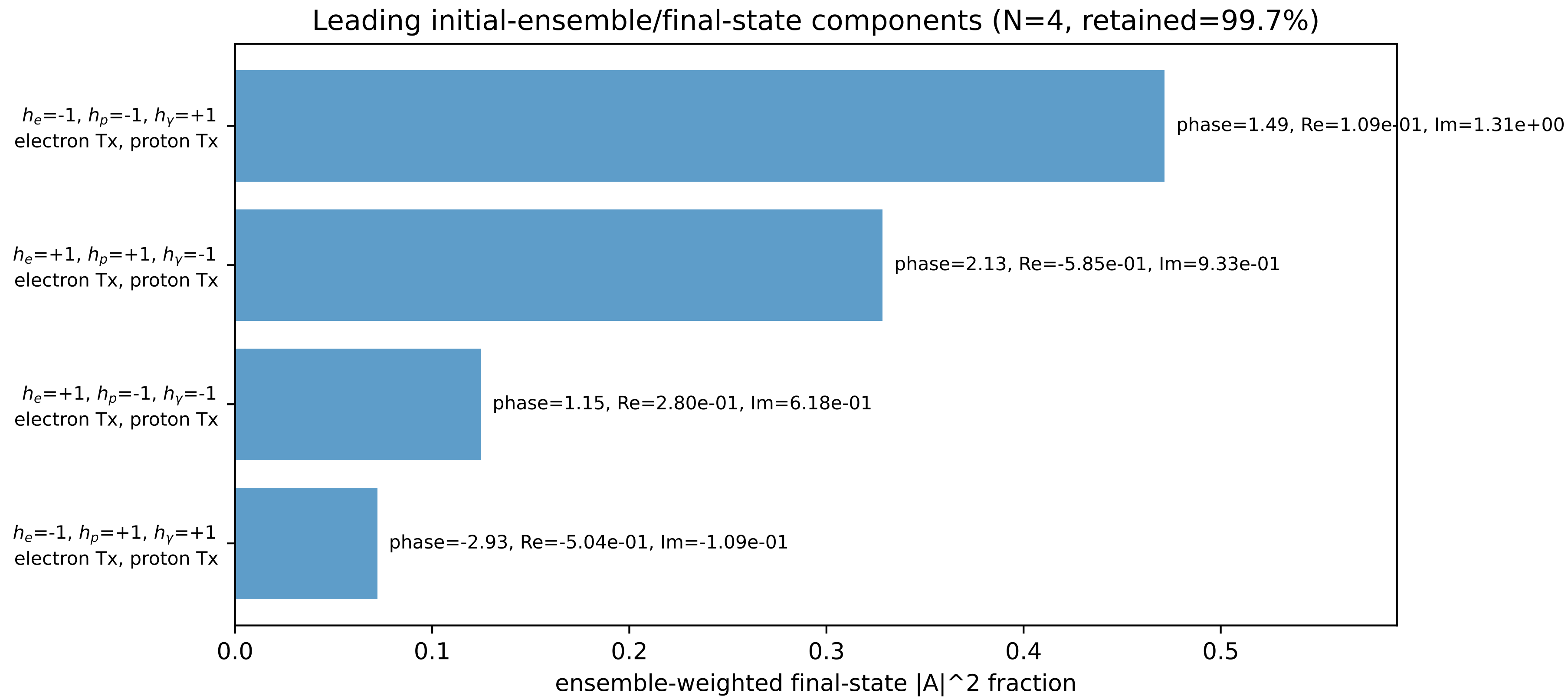


c_mu_gamma_mid_egamma_txtx_region_2 C_{mu\gamma}=0.724221
region: mid_Egamma_TxTx_region_2 final-pair delta_xy=1.87462 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.69401
θ_{in}=1.57079, ϕ_ℓ=5.98866, ϕ_P=2.84707
E_γ=1, ϕ_γ=5.98866
Q^2=9.8142, x_B=0.668979, t=-5.39839, W^2=5.73606, y=0.711298
$\theta(\ell', \gamma)$=107.408 deg

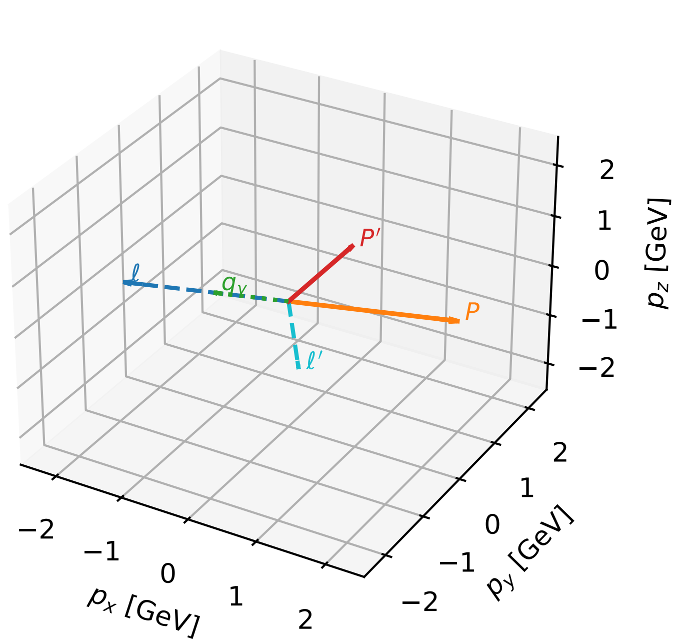
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 2.1274 p_y= -0.6453 p_z= -0.0000
P E= 2.4129 p_x= -2.1274 p_y= 0.6453 p_z= 0.0000
ℓ' E= 1.7022 p_x= -0.9569 p_y= -1.4037 p_z= 0.0000
P' E= 1.9364 p_x= 0.0000 p_y= 1.6940 p_z= 0.0000
q_γ E= 1.0000 p_x= 0.9569 p_y= -0.2903 p_z= 0.0000





Tx muon + Tx proton characteristic max $C_{\mu\gamma}$ configuration

3D momenta



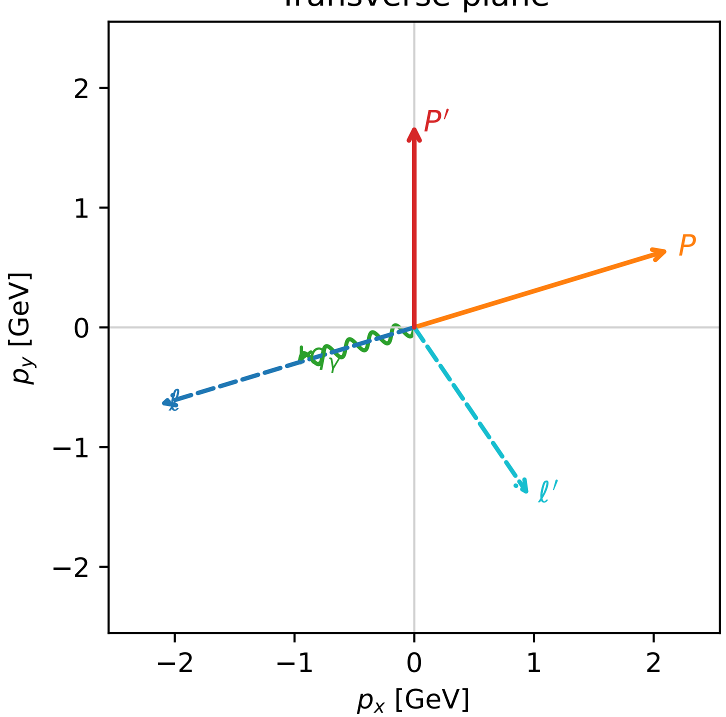

```
c_mu_gamma_mid_egamma_txtx_region_3  C_{mu\gamma}=0.724221
region: mid_Egamma_TxTx_region_3  final-pair delta_xy=1.87462 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.69401
θ_{in}=1.57079, ϕ_ℓ=3.43612, ϕ_P=0.294524
E_γ=1, ϕ_γ=3.43612
Q^2=9.8142, x_B=0.668979, t=-5.39839, W^2=5.73606, y=0.711298
$\theta(\ell',\gamma)$=107.408 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -2.1274 p_y= -0.6453 p_z= -0.0000
P E= 2.4129 p_x= 2.1274 p_y= 0.6453 p_z= 0.0000
ℓ' E= 1.7022 p_x= 0.9569 p_y= -1.4037 p_z= 0.0000
P' E= 1.9364 p_x= 0.0000 p_y= 1.6940 p_z= 0.0000
q_γ E= 1.0000 p_x= -0.9569 p_y= -0.2903 p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=99.7%)